Classical vs Permutation Testing for Relative Mobility Analysis

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12/9

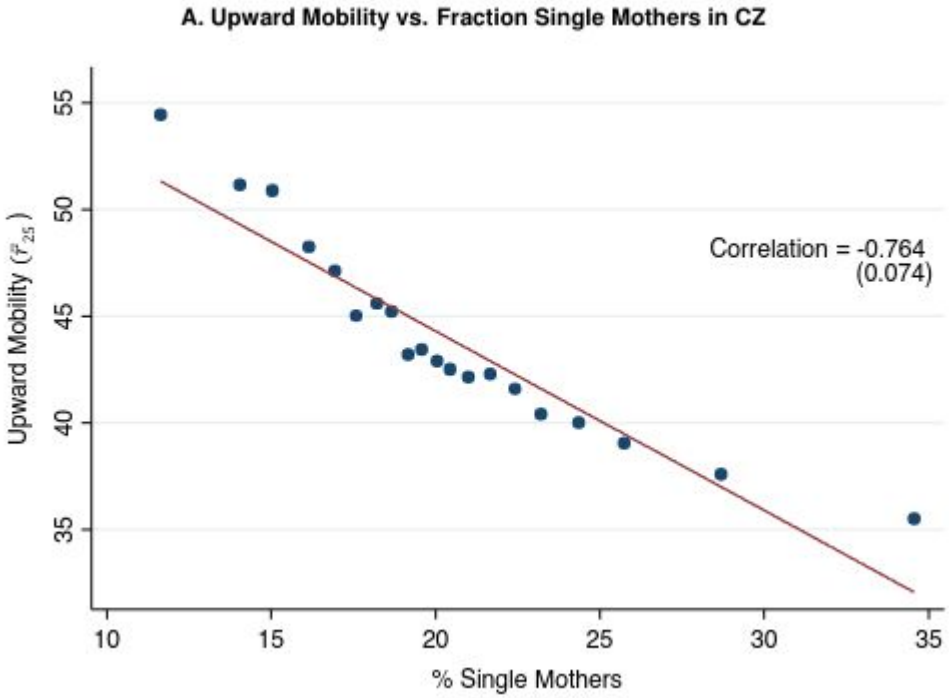
Why This Paper Matters

Policy-Relevant Findings:

- Identified single-parent households as strongest predictor

- Linked segregation, inequality, social capital to children's outcomes

"The fraction of children living in single-parent households is the single strongest correlate of upward income mobility among all the variables we explored" (p. 40)



ONLINE APPENDIX FIGURE XIIa

What Is Relative Mobility?

The Dependent Variable

- Rank-rank slope: correlation between parent income rank and child income rank

- Higher value = less mobility (children's outcomes more tied to parents')

- National average: 0.34

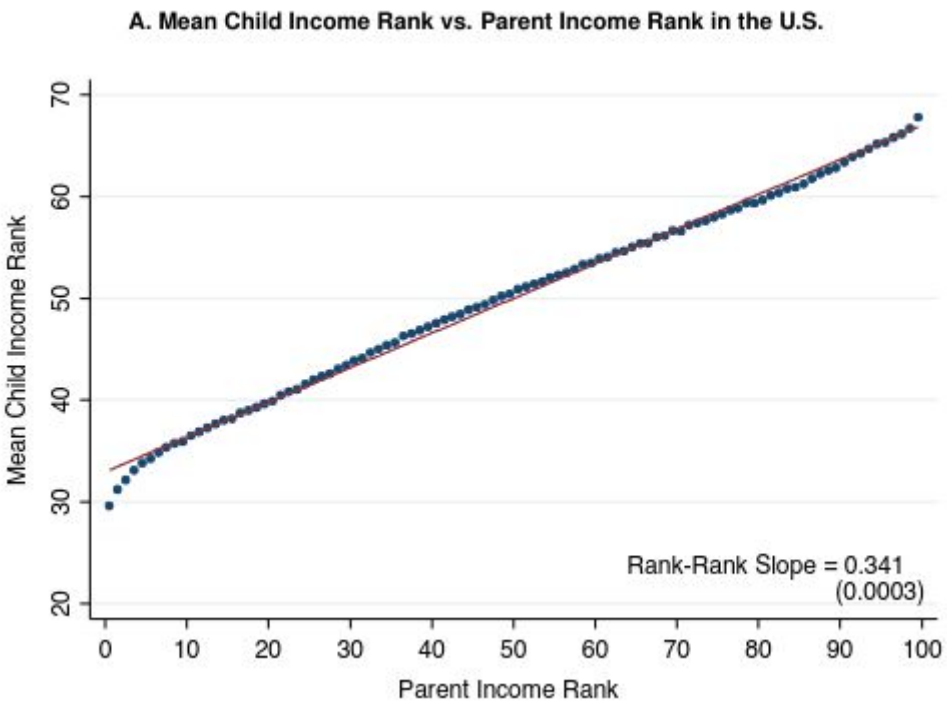


FIGURE II: Association between Children's and Parents' Percentile Ranks

Sources

Online Data Table 5 provides the *dependent* variable: relative mobility estimates for 741 Commuting Zones (CZs). These estimates were computed from IRS tax records covering over 40 million children born between 1980-1991, linked to their parents' tax returns. (Table VIII only used the 1980-82 cohorts)

Online Data Table 8 provides the *independent* variables: 35 community-level characteristics for each CZ, including measures of segregation, income inequality, school quality, social capital, family structure, and local labor market conditions.

Link:
https://opportunityinsights.org/data/?geographic_level=0&topic=0&paper_id=592#resource-listing

Why Did I Replicate Table VIII From The Paper?

- Represents the core contribution of Chetty et al. (2014): **identifying which community-level characteristics are associated with intergenerational mobility.**

- Section VI of the paper (p. 31): "we correlate our measures of intergenerational mobility with local area characteristics"

  In basic terms:

  - Table 5 (online data) = A list of neighborhoods and how easy it is for kids there to do better than their parents (the "mobility score")

  - Table 8 (online data) = A list of the same neighborhoods and facts about them (like: how many single moms? how good are the schools? how much crime?)

  - Table VIII (from paper) = The answer to: "Do neighborhoods with more __ have kids who do better or worse?"

  Table 5 (data) + Table 8 (data) → R code → Recreates RM results in Table VIII

Why Permutation Tests?

Classical OLS assumes:

1. Normal residuals - Errors follow normal distribution
2. Homoscedasticity (*ho-mo-ske-das-tis-i-tee*) - Constant error variance across X
3. Large sample approximations - t-test relies on asymptotic theory

Permutation tests:

1. No distributional assumptions - Uses data itself for null distribution
2. Exact p-values under the null - Under exchangeability, p-values are exact
3. Robust to violations - Less sensitive to violations (with proper studentization)

Methodology

Non-Permutation Methods (4):

Method	Properties	Test Statistic
summary(lm)	t = beta/SE(beta)	Assumes normality
cor.test	Pearson r	Classical t-test for correlation
waldtest HC0	Wald F	Heteroskedasticity-robust
waldtest HC3	Wald F	HC-robust + small-sample correction

Permutation Methods (2):

Method	Properties	Test Statistic
perk	Pearson r	Simple permutation correlation
DiCiccio & Romano	Robust Wald S□	HC-robust, studentized (Theorem 3.2)

Why Multiple Methods?

If…

All methods agree → Findings are robust

Classical != Permutation → Normality assumption issue

perk != DiCiccio & Romano → Heteroskedasticity issue

HC0 != HC3 → High-leverage points affecting inference

The goal is to test whether Chetty et al.'s findings are robust to ALL these concerns

Covariates Tested

35 covariates across 10 categories

1. Race (1)
2. Segregation (5)
3. Income Distribution (5)
4. K-12 Education (4)
5. Social Capital (3)
6. Family Structure (3)
7. Tax (4)
8. College (3)
9. Local Labor Market (4)
10. Migration (3)

ONLINE APPENDIX TABLE VIII
Correlates of Intergenerational Mobility Across Commuting Zones

	Dep. Var.:	Absolute Upward Mobility										Relative Mobility	
		Baseline		State FEs		Pop. Weighted		Urban Areas Only		Controls			
		(1)		(2)		(3)		(4)		(5)		(6)	
Race	Fraction Black Residents	-0.580	(0.066)	-0.353	(0.048)	-0.616	(0.074)	-0.673	(0.063)			0.631	(0.048)
Segregation	Racial Segregation Theil Index	-0.361	(0.045)	-0.274	(0.027)	-0.311	(0.092)	-0.360	(0.068)	-0.273	(0.046)	0.406	(0.048)
	Income Segregation Theil Index	-0.393	(0.065)	-0.260	(0.036)	-0.169	(0.105)	-0.184	(0.068)	-0.267	(0.054)	0.183	(0.063)
	Segregation of Poverty (<p25)	-0.407	(0.066)	-0.261	(0.038)	-0.216	(0.098)	-0.210	(0.066)	-0.274	(0.054)	0.218	(0.059)
	Segregation of Affluence (>p75)	-0.369	(0.064)	-0.250	(0.035)	-0.142	(0.106)	-0.155	(0.070)	-0.250	(0.052)	0.146	(0.063)
	Share with Commute < 15 Mins	0.605	(0.126)	0.342	(0.092)	0.335	(0.115)	0.548	(0.080)	0.415	(0.131)	-0.447	(0.074)
Income Distribution	Household Income per Capita for Working-Age Adults	0.050	(0.071)	-0.013	(0.075)	0.046	(0.092)	0.043	(0.076)	0.064	(0.080)	-0.145	(0.081)
	Gini coefficient for Parent Income	-0.578	(0.093)	-0.281	(0.050)	-0.236	(0.162)	-0.537	(0.120)	-0.362	(0.086)	0.346	(0.089)
	Top 1% Income Share for Parents	-0.190	(0.072)	-0.065	(0.031)	0.059	(0.094)	-0.144	(0.069)	-0.072	(0.065)	0.019	(0.063)
	Gini Bottom 99%	-0.647	(0.092)	-0.433	(0.063)	-0.416	(0.123)	-0.616	(0.114)	-0.470	(0.104)	0.473	(0.090)
	Fraction Middle Class (Between National p25 and p75)	0.679	(0.111)	0.500	(0.102)	0.293	(0.129)	0.551	(0.126)	0.458	(0.145)	-0.451	(0.109)
K-12 Education	School Expenditure per Student	0.246	(0.095)	0.026	(0.099)	0.219	(0.088)	0.236	(0.092)	0.053	(0.082)	-0.279	(0.092)
	Teacher-Student Ratio	-0.328	(0.100)	-0.213	(0.128)	0.062	(0.139)	0.024	(0.104)	-0.249	(0.088)	0.009	(0.108)
	Test Score Percentile (Controlling for Parent Income)	0.588	(0.087)	0.466	(0.074)	0.176	(0.220)	0.413	(0.147)	0.393	(0.093)	-0.317	(0.122)
	High School Dropout Rate (Controlling for Parent Income)	-0.574	(0.089)	-0.413	(0.060)	-0.433	(0.100)	-0.441	(0.108)	-0.440	(0.086)	0.328	(0.099)
Social Capital	Social Capital Index (Rupasingha and Goetz 2008)	0.641	(0.091)	0.349	(0.092)	0.299	(0.131)	0.517	(0.116)	0.478	(0.097)	-0.327	(0.085)
	Fraction Religious	0.521	(0.085)	0.357	(0.061)	0.410	(0.096)	0.417	(0.096)	0.484	(0.065)	-0.101	(0.090)
	Violent Crime Rate	-0.380	(0.146)	-0.163	(0.058)	-0.149	(0.166)	-0.367	(0.145)	-0.244	(0.062)	0.217	(0.140)
Family Structure	Fraction of Children with Single Mothers	-0.764	(0.074)	-0.571	(0.085)	-0.613	(0.129)	-0.719	(0.063)	-0.611	(0.066)	0.641	(0.046)
	Fraction of Adults Divorced	-0.486	(0.100)	-0.333	(0.085)	-0.389	(0.074)	-0.346	(0.103)	-0.569	(0.086)	0.158	(0.088)
	Fraction of Adults Married	0.571	(0.062)	0.417	(0.063)	0.221	(0.127)	0.377	(0.069)	0.365	(0.089)	-0.370	(0.078)
Tax	Local Tax Rate	0.325	(0.070)	0.135	(0.073)	0.155	(0.092)	0.182	(0.073)	0.207	(0.071)	-0.328	(0.061)
	Local Government Expenditures per Capita	0.186	(0.083)	0.074	(0.028)	0.192	(0.087)	0.085	(0.079)	0.107	(0.083)	-0.301	(0.080)
	State EITC Exposure	0.245	(0.064)			0.279	(0.076)	0.355	(0.073)	0.163	(0.073)	-0.144	(0.047)
	State Income Tax Progressivity	0.207	(0.146)			0.265	(0.070)	0.198	(0.098)	0.155	(0.133)	-0.150	(0.106)
College	Number of Colleges per Capita	0.200	(0.114)	-0.015	(0.118)	0.108	(0.088)	-0.045	(0.076)	0.060	(0.142)	-0.125	(0.052)
	Mean College Tuition	-0.018	(0.067)	-0.044	(0.039)	0.058	(0.096)	-0.015	(0.087)	-0.029	(0.066)	0.109	(0.064)
	College Graduation Rate (Controlling for Parent Income)	0.155	(0.062)	0.141	(0.052)	0.107	(0.089)	0.120	(0.095)	0.173	(0.073)	-0.025	(0.057)
Local Labor Market	Labor Force Participation Rate	0.212	(0.086)	-0.045	(0.052)	0.022	(0.090)	0.267	(0.113)	0.146	(0.073)	-0.237	(0.082)
	Fraction Working in Manufacturing	-0.261	(0.091)	0.007	(0.079)	-0.158	(0.090)	-0.128	(0.096)	0.002	(0.085)	0.393	(0.070)
	Growth in Chinese Imports 1990-2000 (Autor and Dorn 2013)	-0.175	(0.078)	0.006	(0.023)	0.001	(0.070)	0.008	(0.102)	-0.107	(0.048)	0.171	(0.083)
	Teenage (14-16) Labor Force Participation Rate	0.631	(0.087)	0.358	(0.098)	0.299	(0.153)	0.540	(0.109)	0.388	(0.090)	-0.516	(0.084)
Migration	Migration Inflow Rate	-0.258	(0.074)	-0.186	(0.049)	-0.146	(0.076)	-0.040	(0.078)	-0.285	(0.069)	-0.084	(0.067)
	Migration Outflow Rate	-0.163	(0.070)	-0.162	(0.048)	0.062	(0.094)	0.013	(0.076)	-0.145	(0.071)	-0.150	(0.070)
	Fraction of Foreign Born Residents	-0.027	(0.064)	-0.014	(0.039)	0.237	(0.083)	0.092	(0.064)	-0.004	(0.051)	-0.247	(0.055)

Code Flow - Beta Replication

Step 1: Replicate the original coefficients

1. Load Table 5 (mobility) and Table 8 (covariates)
2. Merge by Commuting Zone (CZ)
3. Filter to CZs with n >= 250 children
4. For each covariate:
   a. Standardize both X and Y (mean=0, SD=1)
   b. Run OLS: Y = alpha + beta*X + error
   c. Standardized beta = Pearson correlation
5. Compare to Table VIII Column 6 (Relative Mobility)

Result: All 35 betas match exactly (difference = 0)

Beta Replication Results

Covariate	Paper beta	Replication beta	Beta diff
Fraction Black Residents	0.631	0.631	0
Racial Segregation	0.406	0.406	0
Income Segregation	0.183	0.183	0
Segregation of Poverty (<p25)	0.218	0.218	0
Segregation of Affluence (>p75)	0.146	0.146	0
Share with Commute < 15 Mins	-0.447	-0.447	0
Household Income per Capita	-0.145	-0.145	0
Gini Coefficient	0.346	0.346	0
Top 1% Income Share	0.019	0.019	0
Gini Bottom 99%	0.473	0.473	0
Fraction Middle Class	-0.451	-0.451	0
School Expenditure per Student	-0.279	-0.279	0
Teacher-Student Ratio	0.009	0.009	0
Test Score Percentile (Inc. Adj.)	-0.317	-0.317	0
HS Dropout Rate (Inc. Adj.)	0.328	0.328	0
Social Capital Index	-0.327	-0.327	0
Fraction Religious	-0.101	-0.101	0
Violent Crime Rate	0.217	0.217	0
Fraction Single Mothers	0.641	0.641	0
Fraction Divorced	0.158	0.158	0
Fraction Married	-0.37	-0.37	0
Local Tax Rate	-0.328	-0.328	0
Local Gov Expenditures	-0.301	-0.301	0
State EITC Exposure	-0.144	-0.144	0
State Income Tax Progressivity	-0.15	-0.15	0
Colleges per Capita	-0.125	-0.125	0
Mean College Tuition	0.109	0.109	0
College Grad Rate (Inc. Adj.)	-0.025	-0.025	0
Labor Force Participation	-0.237	-0.237	0
Manufacturing Employment Share	0.393	0.393	0
Growth in Chinese Imports	0.171	0.171	0
Teen Labor Force Participation	-0.516	-0.516	0
Migration Inflow Rate	-0.084	-0.084	0
Migration Outflow Rate	-0.15	-0.15	0
Fraction Foreign Born	-0.247	-0.247	0

Code Flow - Significance Testing

Step 2: Compare inference methods

For each of 35 covariates:

1. Compute 4 non-permutation p-values:
   - summary(lm) —> p_classical
   - cor.test —> p_cor_test
   - waldtest(HC0) —> p_waldtest_hc0
   - waldtest(HC3) —> p_waldtest_hc3

2. Compute 2 permutation p-values (B = 10,000):
   - perk —> p_perk
   - DiCiccio-Romano —> p_perm_reg

3. Check significance at alpha = 0.05

Compare: Do all 6 methods agree?

Permutation Test Implementation

DiCiccio & Romano (2017) Theorem 3.2:

Test statistic: $$S_n = n\,\beta'\left(\Sigma^{-1}\,\Omega\,\Sigma^{-1}\right)\beta$$

Where:
- beta = OLS coefficient
- Sigma = (1/n) X'X
- Omega = HC0 robust covariance (handles heteroskedasticity)

Procedure:
1. Compute S□ from your actual data
2. Shuffle Y values randomly (break the X-Y relationship)
3. Recompute S□ on shuffled data
4. Repeat 10,000 times → builds the "null distribution"
5. p-value = proportion of shuffled S□ ≥ observed S□

Key: Studentization makes test robust to heteroskedasticity

Main Results

Statistic	Value
Total covariates	35
Significant	32/35 (91.4%)
Non-Significant	3/35 (8.6%)
Methods Agreeing	34/35 (97.1%)

5 of 6 methods agree for all covariates; HC3 disagrees on 1 (Violent Crime Rate)

Coefficient Plot

Strongest Predictors of LESS Mobility:

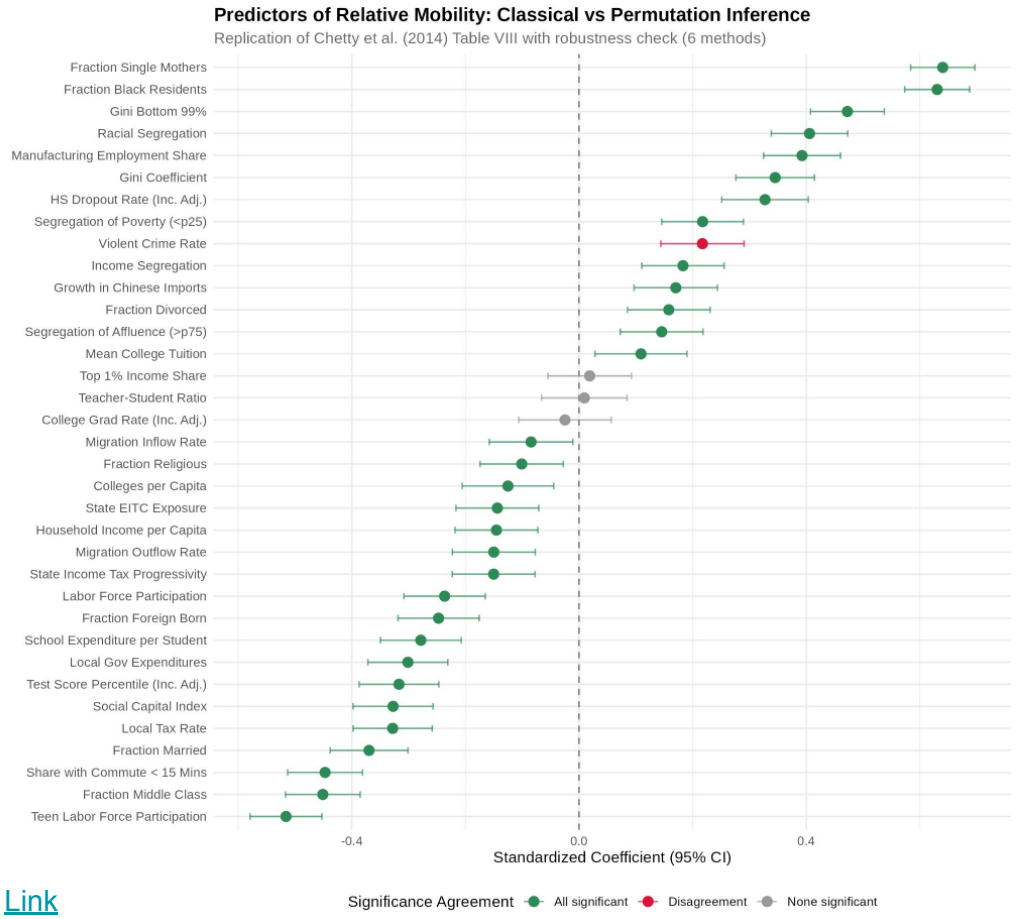
- Fraction Single Mothers
- Fraction Black Residents

No Significance:

- Top 1% Income Share
- Teacher-Student Ratio
- College Grad Rate

Strongest Predictors of MORE Mobility

- Teen Labor Force Participation
- Fraction Middle Class



Link

The One Disagreement - Violent Crime Rate

Method	p-value	Significant?
summary(lm)	<0.001	YES
cor.test	<0.001	YES
waldtest HC0	0.030	YES (barely)
waldtest HC3	0.247	NO
perk	<0.001	YES
perm_reg	<0.001	YES

What's happening:

- HC0 p-value: 0.030 (just under 0.05)
- HC3 p-value: 0.247 (not even close)
- HC3 inflates standard errors for high-leverage observations: $1/(1-h_{ii})^2$
- Some CZs have unusual violent crime rates AND unusual mobility → high leverage

Interpretation: The Violent Crime Rate finding is sensitive to outliers/leverage. The most conservative classical method (HC3) does not find it significant.

Conclusion

Research Question: Do Chetty et al.'s findings depend on classical inference assumptions?

Answer: No.

- 34/35 covariates show identical significance across 6 methods
- Classical OLS, robust standard errors, and permutation tests largely agree
- One exception (Violent Crime Rate) reveals sensitivity to high-leverage observations

Bottom Line: The Land of Opportunity findings are methodologically sound.

References

- Chetty, R., Hendren, N., Kline, P., & Saez, E. (2014). Where is the land of opportunity? The geography of intergenerational mobility in the United States. Quarterly Journal of Economics, 129(4), 1553-1623.
- DiCiccio, C. J., & Romano, J. P. (2017). Robust permutation tests for correlation and regression coefficients. Journal of the American Statistical Association, 112(519), 1211-1220.
- Data: opportunityinsights.org